



**AN ENHANCEMENT OF MULTI-OBJECTIVE WEIGHTED HUNGARIAN
ALGORITHM FOR PRIORITY-BASED RELIEF GOODS
DISTRIBUTION IN DISASTER-PRONE
PHILIPPINE BARANGAYS**

A Thesis Presented to the
Faculty of the Computer Science Department
College of Information Systems and Technology Management
Pamantasan ng Lungsod ng Maynila
(University of the City of Manila)
Intramuros, Manila

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Computer Science (BSCS)

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CONSULTATION RECORDS

	Date	Comments/Suggestions	Adviser Signature
1	March 31, 2026	After reviewing the background of your study, it was observed that the discussion focuses more on the proposed system than on the Hungarian Algorithm itself. The study should primarily discuss the algorithm, including its current applications, advantages, existing limitations—particularly its reliance on a single objective—and how these limitations affect priority-based relief-goods distribution. These discussions must be supported by relevant scholarly studies and credible sources. The proposed system should not be the central focus of the research. Instead, it should serve only as a supporting tool for implementing, demonstrating, testing, and evaluating the Enhanced Multi-Objective Weighted Hungarian Algorithm.	
2	April 24, 2026	After reviewing the study, the following revisions are recommended: 1. Strengthen the discussion of the Hungarian Algorithm by explaining its current applications, advantages, limitations, and the effects of using distance as its only allocation criterion. 2. Explain why the standard Hungarian Algorithm must be enhanced to consider household urgency and resource compatibility in relief-goods distribution. 3. Ensure that each Statement of the	



		Problem addresses a distinct concern. The SOPs may separately focus on the single-objective limitation, dynamic reassignment when household urgency changes, and the comparative performance of the standard and enhanced algorithms. 4. Ensure that the SOPs focus on the algorithmic concerns rather than solely on the features and functions of the proposed system. 5. Clearly explain how household urgency, geographical distance, and resource compatibility will be measured and incorporated into the allocation process. 6. Indicate whether barangay officials, disaster-risk reduction personnel, or other relevant experts will be consulted to validate the criteria and evaluate the allocation results. 7. Compare the standard and enhanced Hungarian Algorithms using consistent performance metrics, such as allocation accuracy, compatibility rate, prioritization efficiency, total assignment cost, and execution time. 8. If applicable, include other relevant assignment or optimization algorithms to further support the selection of the Hungarian Algorithm as the foundation of the study. The researchers are also advised to develop a system for the implementation of the study. However, the primary focus of the research should remain on the Hungarian Algorithm and its proposed enhancement. The system should serve as a supporting tool for demonstrating, testing, and evaluating the standard and enhanced	
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		algorithms.	
3	May 12, 2026	The following revisions and clarifications are recommended: 1. Provide evidence that the dataset was obtained from legitimate and reliable sources, such as official barangay records, disaster-risk reduction offices, or other authorized institutions. If a simulated dataset is used, clearly explain how it was generated and validated to represent realistic household and relief-distribution conditions. 2. The proposed system should display a clear side-by-side comparison between the existing distance-based Hungarian Algorithm and the Enhanced Multi-Objective Weighted Hungarian Algorithm. The comparison should present their respective allocation results and performance metrics, including distance, household urgency, resource compatibility, allocation accuracy, total assignment cost, and execution time.	



4	September 04, 2026	The following revisions and clarifications are recommended: 1. The proposed system should display a clear side-by-side comparison between the existing distance-based Hungarian Algorithm and the Enhanced Multi-Objective Weighted Hungarian Algorithm. The comparison should present their respective allocation results and performance metrics, including distance, household urgency, resource compatibility, allocation accuracy, total assignment cost, and execution time. 2. Incorporate Statements of the Problem 1–3 into the system and ensure that each SOP is supported by a corresponding feature, computation, and output. The results should clearly demonstrate how the enhanced algorithm addresses the limitations of the existing algorithm, implements the weighted cost matrix, and compares runtime performance across different dataset sizes.	
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5	September 09, 2026		
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